Chapter 9 — Debounced Network Search

> **Part II — Networking Patterns** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 1 (Networking Core — `ApiService`, `Outcome` / `AppError`), Ch 2 (Network Reliability — retry/backoff), Ch 6 (UI State & UDF — `StateFlow`, `UiState`), Ch 7 (One-time events). This chapter extends the StreamCart networking core rather than inventing a new client.
>
> **Reference surface in StreamCart:** the **Shop search bar** (product catalog) and the **Watch search bar** (video catalog). We use product search as the running example and note where video search differs.

9.1 Problem Definition and Real-World Context

A search box that queries the server looks trivial — bind a text field, call the API on every keystroke, render results. That naive version is one of the most common performance and correctness failures shipped to production.

Consider a user typing **"wireless headphones"** (18 characters). Keystroke-per- request issues ~18 network calls in under two seconds. The problems compound:

- **Wasted work.** Seventeen of those responses are thrown away the instant the next keystroke arrives. The device burned radio, battery, CPU (JSON parsing), and the backend served 18× the necessary load — multiplied across millions of users this is a real infrastructure cost.
- **Out-of-order responses (the classic race).** The request for `"wireles"` may complete *after* the request for `"wireless"` because of network jitter. Without ordering guarantees the UI renders results for a stale query — the user sees headphone results flicker back to a partial query's results. This is the single most important correctness bug in search.
- **UI jank.** Each response triggers a list diff and recomposition. Firing them at keystroke speed causes dropped frames on the main thread.
- **Rate limiting.** Aggressive clients trip server-side 429s, degrading the experience for the very user who is typing fastest.

The fix is **debouncing** (wait for the user to pause before querying), combined with **cancellation of superseded requests**, **deduplication**, and **single-source-of-truth state**. What seems like a UI detail is in fact a concurrency and data-flow problem — which is exactly why it is an architectural topic and a staple interview question.

Why this is an *architectural* concern, not a coding task

The correctness of search depends on how events flow through the system (keystrokes → debounce → cancellation → request → ordered state update), how concurrency is bounded, and how failures (empty, offline, 429) are modeled as state. Those are design decisions, not syntax.

Three practical use cases

1. **E-commerce product search (StreamCart Shop).** Debounced autocomplete over a large catalog, with recent-search history, category scoping, and suggestions. Latency and relevance drive conversion, so the client must never show stale results.
2. **OTT title search (StreamCart Watch / Netflix-style).** Search across a smaller, cacheable catalog where instant local prefix matching can precede the network call; results include mixed types (titles, people, collections).
3. **Social user/@mention search (Instagram/Twitter-style).** Very high keystroke frequency, strong need to cancel stale queries, and a "no results" state that must be distinguished from "still typing" and "error."

Other domains: fintech (payee/beneficiary lookup with strict input validation), travel (airport/city autocomplete with offline fallback lists), productivity (document/contact search with local + remote merge).

9.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	As the user types, show matching results from the server.
F2	Do not query on every keystroke; wait for a short typing pause (debounce).
F3	Only the **latest** query's results are ever shown (no out-of-order rendering).
F4	Identical consecutive queries (e.g., type then delete back to same text) must not refetch.
F5	Blank/whitespace query clears results and cancels in-flight work.
F6	Show distinct states: idle, typing/loading, results, empty, error.
F7	Minimum query length (e.g., ≥2 chars) before hitting the network.
F8	Survive configuration changes and process death (query text is restored).
F9	Allow retry on error without re-typing.

Non-functional requirements

Attribute	Target / Policy
Performance	Debounce 300 ms; ≤1 in-flight request at a time; no main-thread I/O or parsing.
Scalability	Request volume grows sub-linearly with keystrokes; backend load bounded by debounce + dedup.
Reliability	Transient failures retried (per Ch 2); stale responses never win.
Consistency	UI state always corresponds to the most recently submitted query.
Battery/Network	Cancel superseded requests immediately to free the radio; respect metered networks.
Memory	No leaked coroutines/collectors; bounded result list.
Security	Query strings may be PII (names) — redact in logs (Ch 28).
Accessibility	Announce result count / loading via Compose semantics.

Scope

- **In scope:** debounce, cancellation, dedup, ordered state, error/empty handling, process-death restoration, basic result caching for repeated queries.
- **Out of scope (deferred to other chapters):** full Paging 3 pagination of results (Ch 13), ranking/relevance (backend), local FTS pre-filtering (Ch 35), search *suggestions/autocomplete* UI as a separate concern (Ch 10).

Assumptions & constraints

- A REST endpoint `GET /search/products?q=&limit=` exists (StreamCart backend).
- Server-side ranking is authoritative; the client does not re-rank.
- `minSdk 24`, Kotlin + Coroutines + Flow + Compose, Hilt DI.
- Debounce window and min length are **configurable** (feature flag / remote config, Ch 41) — defaults 300 ms / 2 chars.

Expected behavior (acceptance)

> Typing `headphones` then pausing issues exactly **one** network request for `headphones`; the result list corresponds to `headphones`; deleting to `headphone` and retyping `s` within the debounce window issues no extra call; an offline attempt shows an error state with a working Retry.

9.3 System Design Fundamentals

The solution rests on a small set of concepts, each chosen deliberately:

- **Hot state vs. cold stream.** The query text is *state* (`StateFlow`), the search results are a *reaction* to that state. Modeling the query as a flow lets us apply stream operators (`debounce` , `distinctUntilChanged` , `flatMapLatest`) declaratively.
- **Debounce.** Emit a value only after a quiescent period. `Flow.debounce(ms)` drops intermediate emissions and emits the last one after no new emission for `ms` . This directly implements F2.
- **`flatMapLatest` = cancellation of superseded work.** When a new query arrives, the previous inner flow (the in-flight request) is **cancelled**. This is the operator-level solution to the out-of-order race (F3) and to wasted work.
- **`distinctUntilChanged` = deduplication.** Prevents refetching identical consecutive queries (F4).
- **Structured concurrency.** All work runs in `viewModelScope` , so rotation/ navigation away cancels it deterministically — no leaks (NFR memory).
- **Single Source of Truth.** The `StateFlow<SearchUiState>` is the only thing the UI reads; the repository owns data retrieval and (optional) caching.
- **Backpressure/conflation.** `StateFlow` conflates — the UI always sees the latest state, never a backlog of intermediate loading states.

The elegant insight: **`debounce` + `distinctUntilChanged` + `flatMapLatest`** solve F2, F3, and F4 together in a few declarative lines, and `flatMapLatest` 's cancellation semantics make the out-of-order race *structurally impossible* rather than something we patch with sequence numbers. (We still discuss the manual sequence-number approach in §9.8 for the callback/imperative alternative.)

9.4 High-Level Architecture


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Component responsibilities & ownership

- **`SearchBar`** emits raw text changes; holds no business logic.
- **`SearchViewModel`** owns the `queryFlow` (input state) and derives `uiState` (output state) through the debounce pipeline. It is the **state owner** and the orchestrator of the UDF loop.
- **`SearchProductsUseCase`** encapsulates the single operation "search products for query → `Outcome<List<Product>>` ," keeping the ViewModel free of data concerns and making the rule reusable/testable.
- **`SearchRepository`** is the **SSOT** for search results: it checks the in-memory cache, calls the remote source on a miss, maps DTOs to domain models, and returns a typed `Outcome` .
- **`:core:network`** provides the shared OkHttp/Retrofit stack and the `Outcome` / `AppError` model from Ch 1.

Dependency direction: UI → ViewModel → UseCase → Repository → core. Nothing flows upward except data via suspend/Flow returns. The debounce logic lives in the ViewModel because it is a *presentation timing* concern, not a data concern — the repository must remain a plain request/response boundary so it is reusable by non-debounced callers (e.g., a "search" button, voice search).

9.5 Detailed Component Design


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Component	Input	Output	State	Lifecycle	Errors
`SearchViewModel`	query text, retry/clear events	`StateFlow<SearchUiState>`	`query`, derived `uiState`	`viewModelScope`	maps `AppError` → error state
`SearchProductsUseCase`	query string	`Outcome<List<Product>>`	stateless	caller's coroutine	passes `Outcome` through
`SearchRepositoryImpl`	query string	`Outcome<List<Product>>`	LRU cache	singleton (DI)	wraps exceptions to `AppError`
`ProductSearchApi`	query, limit	`List<ProductDto>`	none	—	throws → caught by repo

Concurrency requirements: the ViewModel pipeline guarantees at most one active inner flow (`flatMapLatest`). The repository must be safe to call from the single active request; its cache access is guarded (see §9.8).

9.6 End-to-End Data Flow

Success path


Success path

Failure path (offline, then retry)


Failure path (offline, then retry)

9.7 Complete Kotlin Implementation

Package: `com.streamcart.search` . Snippets reuse `Outcome` / `AppError` and `DispatcherProvider` from the blueprint conventions (§9 of `BLUEPRINT.md`).

9.7.1 Domain & data models

```
// :domain — domain model consumed by UI
data class Product(
    val id: String,
    val title: String,
    val priceCents: Long,
    val thumbnailUrl: String?,
)

// :data:catalog/remote — network DTO
@Serializable
data class ProductDto(
    @SerialName("id") val id: String,
    @SerialName("title") val title: String,
    @SerialName("price_cents") val priceCents: Long,
    @SerialName("thumbnail_url") val thumbnailUrl: String? = null,
)

fun ProductDto.toDomain() = Product(id, title, priceCents, thumbnailUrl)
```

9.7.2 Remote source (Retrofit)

```
// :data:catalog/remote
interface ProductSearchApi {
    @GET("search/products")
    suspend fun search(
        @Query("q") query: String,
        @Query("limit") limit: Int = 20,
    ): List<ProductDto>
}
```

9.7.3 Repository (SSOT for results + cache)

```
// :data:catalog/repository
interface SearchRepository {
    suspend fun search(query: String): Outcome<List<Product>>
}

@Singleton
class SearchRepositoryImpl @Inject constructor(
    private val api: ProductSearchApi,
    private val dispatchers: DispatcherProvider,
) : SearchRepository {

    // Small LRU cache keyed by normalized query. Guarded by a Mutex because
    // although flatMapLatest serializes the pipeline, other callers
    // (voice search, search button) may hit the same singleton concurrently.
    private val cache = object : LinkedHashMap<String, List<Product>>(
        /* initialCapacity = */ 16, /* loadFactor = */ 0.75f, /* accessOrder = */ true,
    ) {
        override fun removeEldestEntry(eldest: Map.Entry<String, List<Product>>) = size > MAX_ENTRIES
    }
    private val cacheMutex = Mutex()

    override suspend fun search(query: String): Outcome<List<Product>> {
        val key = query.trim().lowercase()

        cacheMutex.withLock { cache[key] }?.let { return Outcome.Success(it) }

        return withContext(dispatchers.io) {
            runCatching { api.search(key) }
                .fold(
                    onSuccess = { dtos ->
                        val products = dtos.map { it.toDomain() }
                        cacheMutex.withLock { cache[key] = products }
                        Outcome.Success(products)
                    },
                    onFailure = { Outcome.Failure(it.toAppError()) },
                )
        }
    }

    private companion object { const val MAX_ENTRIES = 50 }
}
```

> `toAppError()` is the shared mapper from Ch 1 (maps `UnknownHostException → NoConnectivity`, `SocketTimeoutException → Timeout`, `HttpException(429) → RateLimited`, etc.).

9.7.4 Use case

```
// :domain/usecase
class SearchProductsUseCase @Inject constructor(
    private val repository: SearchRepository,
) {
    suspend operator fun invoke(query: String): Outcome<List<Product>> =
        repository.search(query)
}
```

9.7.5 UI state & events

```
// :feature:search/ui
sealed interface SearchUiState {
    data object Idle : SearchUiState                       // below min length / blank
    data object Loading : SearchUiState
    data class Results(val products: List<Product>) : SearchUiState
    data class Empty(val query: String) : SearchUiState    // query ok, zero results
    data class Error(val error: AppError, val retryable: Boolean) : SearchUiState
}
```

9.7.6 ViewModel — the debounce pipeline (the heart of the chapter)

```
// :feature:search/presentation
@HiltViewModel
class SearchViewModel @Inject constructor(
    private val searchProducts: SearchProductsUseCase,
    private val savedState: SavedStateHandle,
) : ViewModel() {

    // Input state. Backed by SavedStateHandle so it survives process death (F8).
    private val query = MutableStateFlow(savedState[KEY_QUERY] ?: "")

    // Manual retrigger channel: lets Retry re-run WITHOUT changing the query
    // (distinctUntilChanged would otherwise swallow a repeat of the same text).
    private val retrigger = MutableSharedFlow<Unit>(extraBufferCapacity = 1)

    @OptIn(ExperimentalCoroutinesApi::class, FlowPreview::class)
    val uiState: StateFlow<SearchUiState> =
        combine(query, retrigger.onStart { emit(Unit) }) { q, _ -> q }
            .debounce { q -> if (q.isBlank()) 0L else DEBOUNCE_MS }   // F2, instant clear on blank
            .distinctUntilChanged()                                   // F4
            .flatMapLatest { raw ->                                   // F3: cancels superseded request
                val q = raw.trim()
                when {
                    q.length < MIN_LENGTH -> flowOf(SearchUiState.Idle)  // F5, F7
                    else -> flow {
                        emit(SearchUiState.Loading)
                        emit(searchProducts(q).toUiState(q))
                    }
                }
            }
            .stateIn(
                scope = viewModelScope,
                started = SharingStarted.WhileSubscribed(5_000),
                initialValue = SearchUiState.Idle,
            )

    fun onQueryChanged(text: String) {
        query.value = text
        savedState[KEY_QUERY] = text
    }

    fun onRetry() { retrigger.tryEmit(Unit) }

    fun onClear() { onQueryChanged("") }

    private fun Outcome<List<Product>>.toUiState(q: String): SearchUiState = when (this) {
        is Outcome.Success -> if (data.isEmpty()) SearchUiState.Empty(q) else SearchUiState.Results(data)
        is Outcome.Failure -> SearchUiState.Error(error, retryable = error.isRetryable())
    }

    private companion object {
        const val DEBOUNCE_MS = 300L
```

```
        const val MIN_LENGTH = 2
        const val KEY_QUERY = "search_query"
    }
}
```

Key decisions, made explicit:

- **`debounce { if blank 0 else 300 }`** — blank input clears instantly (F5) while typed input waits. A fixed `debounce(300)` would delay the "clear" by 300 ms, which feels laggy.
- **`flatMapLatest`** — the previous inner `flow { … }` (including its suspended `searchProducts(q)` call) is cancelled when a new query arrives, giving us cancellation + ordering for free (F3). The `Loading` emit then the result emit model the two UI states of one request.
- **`retrigger` SharedFlow** — Retry must re-issue the *same* query, which `distinctUntilChanged` would otherwise block. Combining a retrigger signal is the idiomatic fix. `onStart { emit(Unit) }` seeds the combine so the pipeline runs on first collection.
- **`WhileSubscribed(5_000)`** — the pipeline stops 5 s after the UI stops collecting (rotation grace window) and restarts when it returns, without re-running a finished request because `StateFlow` replays the last state.
- **`SavedStateHandle`** — restores the query text after process death (F8); the pipeline re-derives results on restore.

9.7.7 Compose UI

```
// :feature:search/ui
@Composable
fun SearchScreen(viewModel: SearchViewModel = hiltViewModel()) {
    val state by viewModel.uiState.collectAsStateWithLifecycle()
    val query by viewModel.queryText.collectAsStateWithLifecycle()  // expose query as StateFlow too

    Column(Modifier.fillMaxSize()) {
        SearchBar(
            query = query,
            onQueryChange = viewModel::onQueryChanged,
            onClear = viewModel::onClear,
        )
        when (val s = state) {
            SearchUiState.Idle -> IdleHint()
            SearchUiState.Loading -> LoadingIndicator(Modifier.semantics { contentDescription = "Searching" })
            is SearchUiState.Results -> ResultsList(s.products)
            is SearchUiState.Empty -> EmptyState(s.query)
            is SearchUiState.Error -> ErrorState(
                error = s.error,
                showRetry = s.retryable,
                onRetry = viewModel::onRetry,
            )
        }
    }
}
```

> `collectAsStateWithLifecycle()` (from `lifecycle-runtime-compose`) stops collection in `STOPPED` and is the correct default over `collectAsState()` for lifecycle safety.

9.7.8 Dependency injection (Hilt)

```
// :data:catalog/di
@Module @InstallIn(SingletonComponent::class)
abstract class SearchDataModule {
    @Binds abstract fun bindSearchRepository(impl: SearchRepositoryImpl): SearchRepository

    companion object {
        @Provides fun provideProductSearchApi(retrofit: Retrofit): ProductSearchApi =
            retrofit.create(ProductSearchApi::class.java)   // Retrofit from :core:network
    }
}
```

9.7.9 Module structure

```
:feature:search
  ui/           SearchScreen, SearchBar, ResultsList, EmptyState, ErrorState
  presentation/ SearchViewModel, SearchUiState
:domain
  usecase/      SearchProductsUseCase
:data:catalog
  repository/   SearchRepository, SearchRepositoryImpl
  remote/       ProductSearchApi, ProductDto
  mapper/       ProductDto.toDomain()
  di/           SearchDataModule
:core:network   Retrofit/OkHttp, Outcome, AppError, toAppError()
```

9.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Rapid keystrokes	Request per key; stale wins	`debounce` + `flatMapLatest` (cancel superseded)
Out-of-order responses	Old query renders over new	`flatMapLatest` cancels the old inner flow before starting the new one — the old response cannot emit into the pipeline
Duplicate query (type/delete)	Redundant fetch	`distinctUntilChanged` + repo LRU cache
Rotation during request	Leaked coroutine / double work	`viewModelScope` + `WhileSubscribed`; result replayed, not refetched
Concurrent repo callers (voice + text)	Cache corruption	`Mutex`-guarded `LinkedHashMap` (LRU)
Process death mid-request	Lost state	`SavedStateHandle` restores query; pipeline re-derives

The out-of-order race, precisely. With `flatMapLatest`, when query *B* supersedes *A*, the operator cancels *A*'s collector. Even if *A*'s HTTP response is already in flight, its continuation resumes into a **cancelled** coroutine and never reaches `emit`. The pipeline is therefore *structurally* single-latest — we do not need sequence numbers.

The imperative alternative (callbacks / RxJava `switchMap` / manual). If you cannot use `flatMapLatest` (e.g., legacy callback API), you implement ordering with a monotonically increasing token:

```
private val seq = AtomicLong(0)

fun onQueryChanged(q: String) {
    val token = seq.incrementAndGet()
    api.searchAsync(q) { result ->
        if (token == seq.get()) render(result)  // drop stale
    }
}
```

This works but is error-prone and does not *cancel* the stale request (wasted network). `flatMapLatest` is strictly better where available. RxJava's `switchMapSingle` is the direct equivalent.

Why not `Mutex` on the pipeline itself? A mutex would serialize requests but *not* cancel stale ones — you would still pay for and possibly render superseded work. Cancellation (`flatMapLatest`) is the right primitive here; `Mutex` is reserved for the shared cache.

9.9 Failure Scenarios and Recovery

For each failure: *how it happens → detection → architectural response → user experience → retry? → recovery → telemetry.*

Failure	Detection	Response	UX	Retry?	Telemetry
No internet	`UnknownHostException` → `NoConnectivity`	`Error(retryable=true)`	"You're offline" + Retry	Yes (manual + auto on reconnect)	`search_error{type=offline}`
Slow network	still within timeout	stay `Loading`	spinner; keep last results if desired	n/a	`search_latency_ms`
Timeout	`SocketTimeoutException` → `Timeout`	`Error(retryable=true)`	"Taking too long" + Retry	Yes	`search_error{type=timeout}`
HTTP 401	interceptor → token refresh (Ch 19)	transparent refresh then replay	invisible	automatic	`auth_refresh`
HTTP 403	`Http(403)`	`Error(retryable=false)`	"Not available in your region"	No	`search_error{type=forbidden}`
HTTP 429	`RateLimited`	backoff (Ch 2); optionally raise debounce	brief wait, then Retry	delayed	`search_error{type=rate_limited}`
HTTP 500	`Http(5xx)`	`Error(retryable=true)` + capped retry	"Something went wrong"	Yes	`search_error{type=server}`
Malformed response	`SerializationException` → `Serialization`	`Error(retryable=false)` ; log sample	generic error	No	`search_error{type=parse}` + schema version
Empty results	success, empty list	`Empty(query)`	"No matches for X" + suggestions	n/a	`search_zero_results{q_len}`
Process death	n/a	`SavedStateHandle` restore	seamless	n/a	—

 9.9 Failure Scenarios and Recovery

> **Design note — "keep last results on error?"** For search, showing a toast/ inline error while *retaining* the last good results often beats blanking the screen. Model this by having the ViewModel hold `lastResults` and emit an `Error` variant that carries them. This is a product decision; we flag it as a trade-off rather than mandating it.

9.10 Scaling the Implementation

- **Level 1 — single screen.** Exactly §9.7: one ViewModel, debounce pipeline, no cache. Correct for an MVP.
- **Level 2 — multi-screen.** Shop and Watch both search. Extract `SearchRepository<T>` or a generic `Searcher` and reuse the pipeline via a shared `DebouncedSearchDelegate` so each feature ViewModel composes it.
- **Level 3 — production.** Add LRU result cache (shown), recent-search history (DataStore, Ch 4), min-length + remote-config debounce (Ch 41), retry/backoff (Ch 2), "keep last results" UX, analytics (Ch 82).
- **Level 4 — large-scale.** Local **FTS** pre-filter (Ch 35) to show instant offline matches before the network responds (merge local + remote), Paging 3 for long result sets (Ch 13), request prioritization (Ch 14), prefetch of likely next queries.
- **Level 5 — enterprise.** A unified **search platform module** serving many feature teams: pluggable sources (products, videos, people), typed result facets, instrumentation baked in, A/B-tested debounce/min-length, and a contract that tolerates backend schema evolution (Ch 16). Governance: one owning team, versioned APIs, backward compatibility across app versions.

Avoid overengineering: a Level-1/2 app does not need FTS or a search platform. Add a layer only when a concrete requirement (offline search, result pagination, multiple teams) demands it.

9.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When to pick
Pipeline operator	`flatMapLatest` (cancel)	`flatMapMerge` / `flatMapConcat`	Always `flatMapLatest` for search; merge/concat only if every query's results matter
Debounce location	ViewModel (Flow)	UI timer / `LaunchedEffect`	ViewModel — testable, survives recomposition
State holder	`StateFlow`	`LiveData`	`StateFlow` (operators, multiplatform, conflation); LiveData only in legacy
Ordering guarantee	`flatMapLatest`	sequence token	Operator for coroutines; token for callbacks
Result source	Remote only	Local FTS + remote merge	Merge when offline search / instant feel required
Cache	In-memory LRU	Room-backed	LRU for ephemeral search; Room if results must persist/offline
Debounce value	300 ms	150 ms / 500 ms	300 is a good default; tune via analytics + A/B (shorter feels snappier but costs calls)
Retrigger for retry	`SharedFlow` combine	re-set query to "" then back	SharedFlow — clean, no UI flicker

Debounce vs throttle. Debounce waits for a pause (ideal for "user finished typing"); throttle caps rate (ideal for continuous streams like scroll). Search is a debounce problem.

9.12 Security Considerations

- **Query strings are often PII** (names, emails for people search). Redact query text in logs/analytics or hash it; never log full queries at INFO (Ch 28).
- **Transport:** HTTPS + certificate pinning (Ch 23); query params are in the URL, so they appear in server access logs — prefer not to send secrets as query strings.
- **Injection:** the client passes `q` verbatim; the **backend** must treat it as untrusted (no SQL/command injection). This is a *backend* guarantee — the client cannot enforce it. State this explicitly in interviews.
- **Autocomplete leakage:** suggestions can reveal other users' data if the backend is careless; the client should not cache another account's suggestions across account switches (Ch 42).
- **Rate-limit abuse:** client debounce reduces load but is **not** a security control — a malicious client bypasses it, so the backend must rate-limit independently.

Client vs backend split: client = debounce, dedup, transport security, log redaction. Backend = authz, injection safety, rate limiting, result filtering by entitlement.

9.13 Performance and Resource Optimization

- **Main thread:** all I/O and JSON parsing on `dispatchers.io`; only state emission reaches Main. Verify with StrictMode (`detectNetwork()`).
- **Network/battery:** `flatMapLatest` cancels superseded requests, freeing the radio promptly; debounce bounds request count. On metered networks consider a longer debounce (Ch 91).
- **Recomposition:** emit *distinct* states; use stable `Product` (data class), `key`-ed `LazyColumn` items, and avoid passing unstable lambdas. Conflated `StateFlow` prevents loading/stale thrash.
- **Memory:** bounded LRU (50 entries); `viewModelScope` cancellation prevents collector leaks; `WhileSubscribed` stops idle pipelines.
- **Profiling:** Android Studio Network Inspector (count requests per typed word — expect 1), `systrace`/Perfetto for jank, Layout Inspector recomposition counts, `adb shell dumpsys gfxinfo` for frame times.

9.14 Testing Strategy

What to fake vs. real: **fake** the repository/API (deterministic latency & errors); use **real** Flow operators and a `TestDispatcher` with virtual time so `debounce` is tested without real delays.

ViewModel debounce + cancellation (virtual time)

```
@OptIn(ExperimentalCoroutinesApi::class)
class SearchViewModelTest {

    private val dispatcher = StandardTestDispatcher()
    private lateinit var useCase: FakeSearchUseCase

    @Before fun setUp() { Dispatchers.setMain(dispatcher) }
    @After fun tearDown() { Dispatchers.resetMain() }

    @Test
    fun `debounces rapid input to a single search`() = runTest(dispatcher) {
        useCase = FakeSearchUseCase(latencyMs = 0)
        val vm = SearchViewModel(useCase, SavedStateHandle())

        vm.uiState.test {   // Turbine
            assertEquals(SearchUiState.Idle, awaitItem())

            vm.onQueryChanged("h")
            vm.onQueryChanged("he")
            vm.onQueryChanged("headphones")   // all within debounce window
            advanceTimeBy(299); expectNoEvents()
            advanceTimeBy(1)                  // cross 300ms

            assertEquals(SearchUiState.Loading, awaitItem())
            val results = awaitItem() as SearchUiState.Results
            assertEquals(1, useCase.callCount)                // F2: one call
            assertEquals("headphones", useCase.lastQuery)     // F3: latest wins
        }
    }

    @Test
    fun `stale response never overrides latest query`() = runTest(dispatcher) {
        // slow "abc", fast "xyz": flatMapLatest must drop "abc"
        useCase = FakeSearchUseCase(latencyFor = mapOf("abc" to 1000L, "xyz" to 10L))
        val vm = SearchViewModel(useCase, SavedStateHandle())
        vm.uiState.test {
            awaitItem() // Idle
            vm.onQueryChanged("abc"); advanceTimeBy(300)
            awaitItem() // Loading(abc)
            vm.onQueryChanged("xyz"); advanceTimeBy(300)
            awaitItem() // Loading(xyz)
            val r = awaitItem() as SearchUiState.Results
            assertEquals("xyz", r.products.first().title)  // abc's 1000ms result dropped
            cancelAndIgnoreRemainingEvents()
        }
    }
}
```

Repository (MockWebServer) + dedup/cache

```
@Test fun `identical query served from cache without second request`() = runTest {
    server.enqueue(MockResponse().setBody(productsJson))
    val repo = SearchRepositoryImpl(api, TestDispatchers(dispatcher))
    repo.search("shoes"); repo.search("shoes")
    assertEquals(1, server.requestCount)      // F4
}

@Test fun `offline maps to NoConnectivity`() = runTest {
    val repo = SearchRepositoryImpl(throwingApi(UnknownHostException()), ...)
    val out = repo.search("shoes")
    assertEquals(AppError.NoConnectivity, (out as Outcome.Failure).error)
}
```

Also cover: min-length (no call below 2 chars), blank clears instantly, Retry re-issues same query, empty list → `Empty`, 429 → backoff path, and a Compose UI test asserting Loading→Results and Retry button visibility on error.

9.15 Production Observability

- **Structured logs** (`:core:analytics`): `search_submitted{q_len, source}` , `search_result{q_len, count, latency_ms, cache_hit}` , `search_error{type, http_code, correlation_id}` . **Never** log raw query text at default levels (PII); log length/hash only.
- **Metrics/KPIs:** p50/p95 search latency, zero-result rate (relevance health), error rate by type, cache hit ratio, avg requests per search session (should be ≈1 after debounce — a spike means debounce regressed), 429 rate.
- **Tracing:** a `correlationId` per submitted query (Ch 84) threaded from the ViewModel through OkHttp header to the backend; crash/ANR breadcrumbs include the current `SearchUiState` type.
- **Debugging in production:** a rising "requests per session" metric is the canary for a broken debounce (e.g., someone recreated the Flow per recomposition); a rising zero-result rate signals an indexing/relevance regression upstream; correlate client latency with backend traces via the ID.
- **Privacy:** honor consent; drop query analytics for users who opt out; never ship raw queries to third-party analytics.

9.16 System Design Interview Preparation

> Debounced search is a favorite because it compresses concurrency, lifecycle, state modeling, and failure handling into one approachable feature. Below: 40+ categorized questions with substantive answers, follow-ups, design curveballs, and a mock walkthrough.

A. Questions & answers

Fundamentals

1. *Why debounce instead of search-on-submit?* Submit is simplest and cheapest but loses the instant-feedback UX users expect; debounce balances responsiveness against load. For very large/expensive queries, submit (or a longer debounce) can be correct.
2. *What debounce value and why 300 ms?* ~300 ms is below the perceptual "laggy" threshold yet long enough to coalesce typing bursts. Tune with analytics/A-B; shorter feels snappier but costs calls.
3. *Difference between debounce and throttle?* Debounce emits after a quiet period (good for "done typing"); throttle emits at a max rate (good for continuous streams).
4. *What is the single biggest correctness bug in naive search?* Out-of-order responses rendering a stale query's results.

5. *Why model query as a `StateFlow`?* It's conflated hot state; lets us apply `debounce` / `distinctUntilChanged` / `flatMapLatest` declaratively and survives recomposition.

Architecture

6. *Where does the debounce belong — UI, ViewModel, or repository?* ViewModel. It's a presentation-timing concern; the repository stays a plain request/response boundary reusable by non-debounced callers.
7. *Why keep a UseCase for a one-line call?* Reuse (button/voice search), testability, and keeping the ViewModel free of data concerns; it's optional in a tiny app (a trade-off, not dogma).
8. *What is the SSOT here?* The `StateFlow<SearchUiState>` for the UI; `SearchRepository` for the result data/cache.
9. *How do Shop and Watch share this?* Extract a generic debounced-search delegate; each feature provides its own repository/result type.

Kotlin / Flow internals

10. *How does `flatMapLatest` prevent the race?* It cancels the previous inner flow's collector when a new upstream value arrives, so the stale request's continuation resumes into a cancelled coroutine and never emits.
11. *Why `distinctUntilChanged`?* Dedup identical consecutive queries (type then delete back).
12. *Why does Retry need a separate signal?* `distinctUntilChanged` would drop a repeat of the same query; a `SharedFlow` retrigger re-runs it.
13. `SharingStarted.WhileSubscribed(5000)` *— why?* Stops the pipeline shortly after the UI stops collecting (survives rotation) and restarts without refetching, since `StateFlow` replays the last value.
14. `collectAsState` *vs* `collectAsStateWithLifecycle`*?* The latter stops collection in STOPPED — correct default to avoid work/leaks while backgrounded.
15. *Why `debounce { if blank 0 }`?* Clear results instantly on blank instead of waiting 300 ms.
16. *Cold vs hot flow in this pipeline?* `query` is hot (StateFlow); the inner `flow { emit(Loading); emit(result) }` is cold, recreated per query.

Concurrency

17. *How many requests are in flight at once?* At most one (flatMapLatest).
18. *Do you need sequence numbers?* Not with `flatMapLatest`; yes with callbacks/imperative APIs.
19. *Is the repository cache thread-safe?* It's guarded by a `Mutex` because multiple callers (text, voice) may hit the singleton concurrently.
20. *Why not a `Mutex` instead of `flatMapLatest`?* Mutex serializes but doesn't cancel stale work; you'd still pay for/possibly render superseded responses.
21. *RxJava equivalent?* `debounce` + `switchMap` (+ `distinctUntilChanged`).

Lifecycle

22. *What cancels in-flight work on navigation away?* `viewModelScope` cancellation.
23. *Process death — what's lost and how restored?* In-memory results lost; `SavedStateHandle` restores query text and the pipeline re-derives results.
24. *Rotation — refetch or not?* No; `WhileSubscribed` + `StateFlow` replay the last state (or you persist results if needed).

Persistence / caching

25. *Where would you cache results and why?* In-memory LRU for ephemeral speed; Room if results must persist/offline. Keyed by normalized query.
26. *Cache invalidation?* Time-based TTL and/or on catalog change events; clear on account switch.
27. *Recent searches?* DataStore list, capped, deduped, cleared on logout.

Networking

28. *How is the networking core reused?* Same OkHttp/Retrofit + `Outcome` / `AppError` from Ch 1; search adds only an endpoint.
29. *Handling 429?* Backoff (Ch 2), optionally raise debounce; backend must rate-limit independently.
30. *401 during search?* Transparent token refresh (Ch 19) then replay.

Scalability

31. *10M users — what changes on the client?* Little on the client; debounce + dedup already bound load. The scaling is backend (index, autocomplete service); client may add FTS pre-filter and A/B-tuned debounce.
32. *How to reduce backend load further?* Longer debounce, min-length, client caching, prefix caching, local FTS, CDN-cached popular queries.

Security

33. *Is query text sensitive?* Often PII; redact/hash in logs; prefer not to send secrets as query params.
34. *Can client debounce be a rate-limit control?* No — trivially bypassed; backend must enforce.
35. *Injection responsibility?* Backend; client sends `q` untrusted.

Performance

36. *How do you keep it off the main thread?* IO dispatcher for calls/parsing; StrictMode to verify.
37. *Recomposition hotspots?* Emit distinct states; stable models; keyed lazy lists.
38. *How verify debounce in prod?* "Requests per search session" metric ≈ 1.

Testing

39. *How test debounce without real delays?* `StandardTestDispatcher` + `advanceTimeBy`; Turbine for emissions.
40. *How test the out-of-order race?* Fake variable latency; assert the slow earlier query's result is dropped (test in §9.14).

Production debugging

41. *Users report "flicker to old results" — diagnose.* Out-of-order race → verify `flatMapLatest` (not `flatMapMerge`) and that the Flow isn't recreated per recomposition.
42. *Backend sees request spikes from one user — cause?* Debounce regressed: Flow rebuilt on each recomposition, or debounce moved to UI with a new timer per keystroke.

C. Deeper follow-ups a Staff/Principal would ask

- "You said `flatMapLatest` cancels the request — does OkHttp actually cancel the socket, or just stop collecting?" (Coroutine cancellation propagates to the suspended Retrofit call, which cancels the OkHttp `Call`; confirm with `Call.isCanceled()` behavior.)
- "If results must be paginated, where does debounce sit relative to Paging 3?" (Debounce the query, then feed it into a `Pager` recreated per query via `flatMapLatest` on the `PagingData` flow.)
- "How would you merge instant local FTS results with slower remote results without flicker?" (Emit local immediately as a partial state, then reconcile with remote; model a `Results(partial=true)` variant.)

D. Design-change challenges (≥5)

1. **"Make it work offline."** Add local FTS (Ch 35); emit local results first, mark remote unavailable; sync index in background (Ch 30).
2. **"Support 10M users / cut backend cost 50%."** Longer/adaptive debounce, min-length, client + edge caching of popular prefixes, local pre-filter.
3. **"Two result sources (products + videos) in one bar."** Generalize to a multi-source searcher; merge with typed facets; each source cancels independently but shares the debounce.
4. **"Backend changed the response schema."** Version the DTO, tolerant parsing with defaults (Ch 16); feature-flag the new path; keep old app versions working.
5. **"Process killed while typing a long query."** `SavedStateHandle` restores text; re-derive; optionally cache last results in Room for instant restore.
6. **"Results must stay visible during errors."** Carry `lastResults` in the error state; show inline banner + Retry instead of blanking.

E. Mock interview walkthrough (30–45 min, abridged)

1. **Clarify (0–5m):** platform, catalog size, offline needed?, result types, latency budget, who owns ranking. → "Remote REST, large catalog, online-first v1, products only, p95 < 400 ms, backend ranks."
2. **Requirements (5–10m):** state F1–F9 and NFRs; call out the out-of-order race as the key risk.
3. **Architecture (10–20m):** draw UI → VM(debounce pipeline) → UseCase → Repo → core; name SSOT and dependency direction; justify debounce in the ViewModel.
4. **Deep dive (20–30m):** the Flow pipeline; explain `flatMapLatest` cancellation solving ordering; cache + dedup; Retry retrigger; process death.
5. **Failure & scale (30–40m):** failure matrix; "keep last results" trade-off; scale to FTS/Paging/multi-source; 10M-users answer (client vs backend split).
6. **Trade-offs & defense (40–45m):** debounce value, LiveData vs StateFlow, StateFlow vs callbacks; respond to objection "why not just search on submit?" with the UX/load trade-off and when submit is actually right.

9.17 Practical Exercises

1. **Beginner.** Implement the debounce pipeline for a single screen; assert via Network Inspector that typing a 10-char word fires exactly one request. *Evaluation:* one request, correct final query, no main-thread I/O.
2. **Intermediate.** Add min-length, instant-clear on blank, and Retry via a retrigger `SharedFlow`. Write a Turbine test proving Retry re-issues the same query. *Evaluation:* tests green; no flicker on retry.
3. **Advanced.** Add an in-memory LRU cache + recent-search history (DataStore); prove dedup with MockWebServer `requestCount`. Then generalize the pipeline into a reusable `DebouncedSearchDelegate<T>` used by two ViewModels. *Evaluation:* zero duplicate requests, delegate reused, no leaks.
4. **Production debugging.** Given a repo where the Flow is rebuilt on every recomposition (requests-per-session ≈ keystrokes), find and fix the bug. *Evaluation:* identifies Flow recreation / wrong collection site; metric returns to ≈1.
5. **System design challenge.** Design offline + online search merging local FTS with remote results without flicker, surviving process death and schema change. *Evaluation:* partial-results state model, cancellation correctness, versioned DTO, background index sync.

(Solutions withheld unless requested.)

9.18 Chapter Summary & Reference Sheet

Key architectural decisions

- Debounce + dedup + cancellation live in the **ViewModel Flow pipeline**; the repository stays a plain request/response SSOT.
- `flatMapLatest` makes the out-of-order race **structurally impossible** — no sequence numbers needed.
- Model search as **exhaustive states** (Idle/Loading/Results/Empty/Error), not booleans.

Important concepts: debounce vs throttle, `flatMapLatest` cancellation, `distinctUntilChanged` dedup, `StateFlow` conflation, `WhileSubscribed`, `SavedStateHandle` restoration, retrigger for retry.

Common mistakes: per-keystroke requests; `flatMapMerge` / `flatMapConcat` instead of `flatMapLatest`; rebuilding the Flow on each recomposition (silent debounce defeat); fixed `debounce` delaying blank-clear; booleans instead of a sealed state; logging raw query text; treating client debounce as a rate-limit control.

Design checklist:

- [ ] debounce (configurable)
- [ ] min-length
- [ ] instant clear
- [ ] `flatMapLatest`
- [ ] `distinctUntilChanged`

- [ ] exhaustive states
- [ ] Retry retrigger
- [ ] `SavedStateHandle`
- [ ] IO off main
- [ ] bounded cache w/mutex
- [ ] redacted logs + correlation ID
- [ ] requests-per-session metric.

Key Kotlin/Flow APIs: `MutableStateFlow`, `debounce`, `distinctUntilChanged`, `flatMapLatest`, `combine`, `stateIn`, `SharingStarted.WhileSubscribed`, `MutableSharedFlow.tryEmit`, `collectAsStateWithLifecycle`, `SavedStateHandle`, `Mutex.withLock`, `StandardTestDispatcher` / `advanceTimeBy`, Turbine `test {}`.

Android/Jetpack: ViewModel, Hilt, Retrofit `@Query`, OkHttp cancellation, DataStore (recent searches), StrictMode (verification).

Official references:

- Kotlin Flow: https://kotlinlang.org/docs/flow.html
- `debounce` / `flatMapLatest` API: https://kotlinlang.org/api/kotlinx.coroutines/kotlinx-coroutines-core/kotlinx.coroutines.flow/
- StateFlow/SharedFlow: https://developer.android.com/kotlin/flow/stateflow-and-sharedflow
- Collect flows safely: https://developer.android.com/topic/architecture/ui-layer#consume-ui-state
- ViewModel + SavedState: https://developer.android.com/topic/libraries/architecture/viewmodel/viewmodel-savedstate
- Testing coroutines: https://developer.android.com/kotlin/coroutines/test

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Chapter review (Phase 4)

- **Technical completeness checklist:** all 18 mandatory sections present; success + failure sequence diagrams; architecture + class diagrams; full Kotlin (models → DI); concurrency race analysis with both operator and imperative solutions; full failure matrix; 5 scaling levels; trade-off tables; security client/backend split; performance + profiling tools; unit/repo/concurrency tests; observability KPIs; 42 interview Q&A + follow-ups
  - 6 design challenges + mock; 5 exercises; reference sheet. ✅
- **Possible missing scenarios to expand later:** multi-source search internals (deferred to a dedicated treatment), voice-search integration, deep Paging 3 integration (Ch 13 cross-ref), accessibility deep dive.
- **Interview coverage:** exceeds the ≥40 bar for a Complex topic.
- **Code review considerations:** expose `queryText` as a `StateFlow` on the ViewModel (referenced by the Compose sample); confirm `toAppError()` from Ch 1 covers 429/401; `FakeSearchUseCase` lives in `:core:testing`.
- **Improvement suggestions:** add a runnable `reference-app/` module later; add an adaptive-debounce (network-aware) variant tied to Ch 91.